



Data Science: Theoretical Practice & Statistical Testing on E-Commerce data

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1 Part I — Practical Analysis: E-Commerce Dataset

1.1 Introduction

This exercise uses the Online Retail “E-Commerce Data”¹ dataset to practise a set of standard statistical tests in Python. The dataset contains invoice level records with information on products, quantities, prices, customers, and countries. You will implement the different hypothesis tests below, interpret the output in plain language, and write a short report explaining what you did and what you learned.

Your work should be organised in a Jupyter notebook and a brief written report of you explaining these tests both in theory and practice.

1.2 Dataset and Basic Preparation

You should begin by:

- Reading the CSV file into a pandas dataframe.
- Restricting attention to rows with positive quantity and positive unit_price.
- Creating a new variable revenue as $\text{revenue} = \text{quantity} \times \text{unit_price}$.

Before running any tests, inspect the data with `head()`, `info()`, and `describe()`, and construct a few basic histograms or boxplots for quantity, unit_price, and revenue. A short description of what the data look like should appear at the start of your report. keep in mind these are the basic first steps on most datasets any further analysis is dependent on the details of data.

1.3 Required Statistical Tests

You should implement the six tests listed below. For each one, the handout suggests a matching test. You are free to justify a different choice if you explain it.

For each test you should:

1. State a clear question in words.
2. Write the null and alternative hypotheses.
3. Choose and run an appropriate test in Python.
4. Report test statistic, p-value, and any relevant summary statistics.
5. Interpret the result in a few sentences in context of the data.

Test 1: Normality of revenue

Question: does the distribution of line item revenue resemble a normal distribution?

Hint for implementation Compute revenue as defined above. Take a random sample of at most a few thousand observations from revenue to avoid unnecessary sensitivity. Use the Shapiro–Wilk test `scipy.stats.shapiro` on this sample. You should also draw a histogram or density plot to support your conclusion.

Hypotheses

- H_0 : the sampled revenue values come from a normal distribution.
- H_1 : the sampled revenue values do not come from a normal distribution.

In your report, describe whether the normality assumption seems plausible and what this implies for later tests that assume normality.

¹<https://www.kaggle.com/datasets/carrie1/ecommerce-data>

Test 2: Association between quantity and unit price

Question: is there an association between quantity ordered and unit price?

Hint for implementation Use the two columns `quantity` and `unit_price` for rows with positive values. Since both variables are numeric and the relationship might not be strictly linear, a Spearman rank correlation is a sensible choice. Use `scipy.stats.spearmanr`.

Hypotheses

- H_0 : there is no monotonic association between quantity and unit price.
- H_1 : there is a monotonic association between quantity and unit price.

Report the correlation coefficient and p-value. Comment on the sign and magnitude of the coefficient and whether the association is weak or strong in practical terms.

Test 3: Difference in revenue between the United Kingdom and other countries

Question: do line item revenues differ on average between the United Kingdom and all other countries combined?

Hint for implementation : use the two-sample t-test

Include the group means, the t statistic, and the p-value in your report. State which group has the higher mean and whether the difference seems large in context.

Test 4: Revenue across top countries

Question: among the top countries by number of transactions, are the mean revenues the same?

Hint for implementation Identify the five countries with the highest number of rows in the dataset. Filter the data to keep only those countries and again use the `revenue` variable. For each country, extract the revenue values and apply a one-way ANOVA using `scipy.stats.f_oneway`.

Report the F statistic, p-value, and the mean revenue for each country. In the discussion, explain which countries appear to have higher or lower typical revenue and how large the differences are.

Test 5: Association between country and high value purchase

Question: is the proportion of high value purchases the same across the main countries?

Report the chi-square statistic, degrees of freedom, and p-value. Comment on whether some countries have a higher share of high value line items than others.

1.4 Expected notebook and report structure

Your Jupyter notebook should be organised into clear sections that mirror the tests above. A suggested structure is:

- Introduction and data loading.
- Basic cleaning and creation of the revenue variable.
- Test 1 through 6 each in separate cells

Each section should contain:

- A short markdown cell stating the question and hypotheses.
- The Python code that performs the test and basic visualisations.

- A short markdown cell that interprets the result in context.

Aim for code that is clean and readable. You do not need to include every intermediate step, but any data filtering or grouping used for a test should be visible in the notebook.

Along with your notebook, write a short report of two to four pages.

- A brief description of the dataset.
- One paragraph per test, summarising the question, the test used, and the main result in words and plots.
- A short closing section that reflects on which tests seemed most appropriate for this dataset and which assumptions might be questionable.

In each test paragraph, mention which variables you used, whether the null hypothesis was rejected at a conventional significance level, and what that tells you about customer or transaction behaviour in this e-commerce setting.

2 Part II — Theoretical Analysis: Study and Performance Dataset

In addition to the coding work on the e-commerce dataset, you will complete a short handwritten assignment on a small study and performance dataset. The aim of this part is to show that you can set up standard hypothesis tests from first principles, work through the formulas by hand, and interpret the results without relying only on Python output.

You should answer these questions on paper. For each test you are asked to follow a fixed structure. Code can be used later to check your results, but the handwritten part will be graded on the clarity and correctness of your statistical reasoning.

2.1 General instructions

For every test described below, you must:

1. State the null and alternative hypotheses clearly in words and using symbols.
2. Write the formula for the test statistic and name the reference distribution under the null hypothesis (for example, t distribution, chi square distribution, F distribution, or normal distribution).
3. Compute all intermediate quantities by hand. This includes means, variances, standard deviations, ranks, contingency table counts, and any other elements that enter the test statistic.
4. Calculate the numerical value of the test statistic using your intermediate results.
5. Explain how the p value is obtained, either by referring to statistical tables or by describing the relevant tail area of the reference distribution.

Solutions that only give a final numeric answer, without showing hypotheses, formulas, intermediate steps, and a decision, or that are not handwritten, will receive no credit for that test.

2.2 Dataset Description

For all six questions in this theoretical section, use the following small dataset. It describes students from three departments, with their weekly study hours, exam scores, and study method.

student_id	hours_studied	exam_score	department	study_method
1	5	55	Engineering	Online
2	8	65	Engineering	InPerson
3	6	60	Engineering	Online
4	10	75	Engineering	InPerson
5	4	50	Business	Online
6	7	68	Business	InPerson
7	3	45	Business	Online
8	9	72	Business	InPerson
9	6	58	Arts	Online
10	11	80	Arts	InPerson
11	2	40	Arts	Online
12	8	70	Arts	InPerson

You may redraw the table in your notes or refer to it as needed. All calculations for this part should be based only on the numbers above.

Normality of exam scores (Shapiro–Wilk)

Question 1

Does the distribution of exam scores in this sample look consistent with a normal distribution?

Task. Consider the twelve values in the `exam_score` column as a single sample. Set up the Shapiro–Wilk test of normality for these data. On paper, you are not expected to compute the full Shapiro–Wilk statistic from scratch, since it involves tabulated coefficients, but you should:

- Write down the null and alternative hypotheses.
- Explain the general form of the Shapiro–Wilk statistic and the reference distribution it uses.
- Compute and report basic descriptive quantities such as the sample mean, sample standard deviation, and a rough check of symmetry (for example, by comparing mean and median).
- Based on these summaries and the idea of the Shapiro–Wilk test, argue whether the normality assumption is reasonable for this small sample.

In your answer you may describe qualitatively what a Q–Q plot or a normality test would show, but you do not need to draw the plot.

Correlation between study hours and exam score (Spearman)

Question 2

Is there an association between hours studied per week and exam score?

Task. Use the pairs (`hours_studied`, `exam_score`) for the twelve students and perform a Spearman rank correlation test by hand.

You should:

- Write the null and alternative hypotheses for the Spearman correlation.
- Rank the hours studied and the exam scores from smallest to largest, assign average ranks if ties occur, and write the two rank sequences clearly.
- Compute the differences in ranks for each student and the sum of squared rank differences.

- Use the usual Spearman formula

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

to obtain the rank correlation coefficient.

- State which reference distribution or approximation would be used to obtain a p value, given the sample size.

Conclude by commenting on the sign and magnitude of r_s and whether the data suggest a monotonic relationship between study hours and exam score.

Exam score difference by study method (two sample t test)

Question 3

Do students using in-person study have different average exam scores from those using online study?

Task. Split the data into two groups based on `study_method`: one group for Online and one for InPerson. Use an independent two sample t test to compare the mean exam scores of these groups.

On paper, you should:

- List the exam scores for Online students and the exam scores for InPerson students separately.
- Write the null and alternative hypotheses for the two sample comparison.
- Compute the sample mean and sample variance for each group.
- Decide whether to assume equal variances or not and write the corresponding t statistic formula.
- Calculate the value of the t statistic and the approximate degrees of freedom.
- Indicate how you would obtain the p value from t tables and whether you would reject the null hypothesis at $\alpha = 0.05$.

Give a short interpretation of what your decision means for the difference in exam performance between the two study methods.

2.3 Exam score across departments (one way ANOVA)

Question 4

Are mean exam scores the same across the three departments (Engineering, Business, Arts)?

Task. Treat department as a three level factor and exam score as the response. Perform a one way analysis of variance by hand.

You should:

- Group the exam scores by department and write each group explicitly.
- State the null and alternative hypotheses for the ANOVA.
- Compute the overall mean of all exam scores.
- Compute the between-group sum of squares and within-group (residual) sum of squares, showing intermediate steps.
- Determine the mean squares for between and within, form the F statistic, and state the numerator and denominator degrees of freedom.
- Explain how you would use F tables to approximate the p value and whether you would reject the null hypothesis at $\alpha = 0.05$.

Summarise whether the data suggest that at least one department has a different mean exam score.

2.4 Department and high performance (chi square test)

Question 5

Is the proportion of high performing students the same across departments?

Task. Define a binary performance variable based on exam score. For this small dataset, a simple choice is to call a student high performing if their exam score is at least 65 and low performing otherwise. Using this definition, construct a contingency table with rows corresponding to departments (Engineering, Business, Arts) and columns corresponding to performance level (high, low).

You should:

- Fill the observed contingency table by counting students in each cell.
- State the null and alternative hypotheses for the chi square test of independence.
- Compute the expected count for each cell under the null hypothesis, based on row and column totals.
- Calculate the chi square test statistic

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

and specify the degrees of freedom.

- Describe how you would use chi square tables to find a p value and compare it with $\alpha = 0.05$.

Write a brief conclusion on whether performance level appears to be independent of department in this sample.

Stay hungry, stay foolish